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Financial distress prediction in emerging markets by using a machine learning approach for financial sustainability

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Abstract

The present study investigates machine-learning approaches to predicting financial distress among Indonesian firms using annual data for 325 companies (2013–2023 3575 observations). The feature set comprises financial and operational predictors, in addition to a binary outcome indicator of distress. In order to mitigate severe class imbalance driven by the rarity of distress events, the Synthetic Minority Over-sampling Technique (SMOTE) is applied within training folds, while the validation and test sets are left untouched in order to avoid leakage. Principal Component Analysis (PCA) facilitates exploratory visualisation. The performance of Support Vector Machine (SVM), Decision Tree, Random Forest, k-Nearest Neighbors (KNN), Gradient Boosting, and a deep neural network (ANN/MLP) is benchmarked under a stratified evaluation protocol. Among conventional models, the Decision Tree demonstrates the strongest F1 performance (0.8980), followed closely by Gradient Boosting (0.8627) and Random Forest (0.8462). In the case of the deep model, the best configuration (50 neurons, Sigmoid activation, 100 epochs, batch size 8) achieved 99.20% accuracy and an F1-score of 0.95, indicating robust minority-class detection alongside high overall correctness. The findings underscore the effectiveness of ensemble learners and optimally tuned deep neural networks in predicting financial distress in the context of emerging markets. The study's practical relevance aligns with the Development Goal (SDG) 8: Decent Work and Economic Growth, the earlier and more reliable detection of financial distress can help firms take preventive actions that reduce job losses and operational disruptions, Allowing businesses to predict and act upon their financial challenges, this study nurtures resilience and sustainable development within the corporate sector, the main driver of economic activities. Besides, addressing financial distress in an emerging market like Emerging Markets contributes to inclusive economic development through stability that helps workers, shareholders, and communities dependent on such businesses.

Keywords Financial distress prediction, Machine learning, Deep learning, Financial risk management, Emerging markets



1 Introduction

Predicting financial distress is a critical area of concern for companies, investors, financial institutions, and regulators, given its significant implications for financial stability and economic well-being. Financial distress is initially accompanied by a company's inability to meet its obligations, both short-term and long-term. This will affect the perceptions of shareholders and creditors, and also impact the broader economy. Therefore, a financial distress prediction method is needed to mitigate early risks before financial distress occurs in a company. In the past, researchers used a number of financial ratios from both balance sheets and income statements, tested using logistic regression. The resulting bankruptcy prediction model is called the Altman Z-Score. Then there's the Ohlson O-Score, which also uses logistic regression. These models have inspired researchers interested in financial distress. Currently, many research models utilize machine learning, such as Support Vector Machines (SVM), k-Neighbors (KNN), Random Forests, and others. These new models offer results that are superior in precision to traditional models.

Deep learning is one of the machine learning methods with the highest accuracy compared to other machine learning models. This method studies vast amounts of data and creates precise patterns for decision-making. Input data includes not only numbers and letters but also images. Studies [1] and [2] concluded that modern machine learning methods offer more precise results than traditional methods. The potential of these modern models depends heavily on the data used for continuous learning and training. A combination of data and news to predict financial distress is also necessary to produce more comprehensive research results, both qualitatively and quantitatively, so that predictions are more precise and reliable [3]. After these results are obtained, feedback from model users is needed to address model deficiencies [4].

A limitation of creating this financial distress model lies in the amount of data used for machine learning processing. Imbalanced data will bias prediction results for minority groups. For example, if distressed individuals are more likely to be sick than healthy individuals in a data set, balanced data is recommended. The use of Synthetic Minority Oversampling Technique (SMOTE) can address this imbalanced dataset. Additionally, features of Principal Component Analysis (PCA) can be used to improve prediction performance. Based on the initial presentation, this study aims to develop a predictive model for financial distress in Indonesian companies using several ratios from financial reports, the number of directors, and macroeconomic variables such as the corruption index in Indonesia. This study attempts to incorporate several industrial sectors, excluding the financial sector, such as banking and insurance. Machine learning methods are used to develop a more robust predictive model, a technique widely used in recent research.

2 Literature review

Financial distress prediction is a very important research topic, especially in Indonesia where economic fluctuations and market dynamics are extraordinary challenges. The study entitled "Comparison of Machine Learning Algorithms in Predicting Financial Distress of Indonesian Companies" aims to explore algorithms, machine learning, and their capabilities in predicting financial distress, where financial distress is a condition that can lead to extraordinary bankruptcy if not addressed in a timely manner. This

literature review discusses existing research on financial distress prediction, focusing on machine learning methodologies, financial ratios, and the unique scope of the Indonesian economy. Machine learning has also emerged as a very powerful tool in financial distress prediction, offering very sophisticated analytical capabilities compared to existing traditional statistical methods. Ratih et al. [5] Highlighted the integration of machine learning and deep learning models with Altman Z-score to predict financial distress in companies listed on the Indonesia Stock Exchange (IDX). This study focuses on the potential of artificial intelligence to improve prediction accuracy, most importantly when comparing the K-Nearest Neighbor and Random Forest models through deep learning approaches such as Convolutional Neural Networks (CNNs). These results demonstrate that machine learning can significantly improve the predictive performance of financial distress models, crucial for stakeholders in decision-making.

In addition to machine learning, financial ratios remain a cornerstone in assessing a company's financial health [6]. Larger companies with increasing liabilities are particularly vulnerable to financial distress, consistent with other studies that suggest liquidity and leverage ratios are crucial for predicting financial distress [7, 8]. The combination of financial ratios and machine learning algorithms can provide comprehensive results in predicting financial distress. The determinants of financial distress can be analyzed across multiple factors, including both financial ratios and external economic factors [9]. The predictive power of specific financial ratios can vary across sectors. This highlights the need for sector-specific models capable of accurately capturing financial distress predictions across industries.

It is crucial to conduct a literature review to better understand the determinants of financial distress [10]. This approach is crucial in developing a robust predictive model, as it allows researchers to identify the most significant variables influencing financial distress under different conditions. These determinants can provide accurate prediction results using machine learning methods, as the algorithm can capture complex relationships that cannot be predicted by traditional methods. This machine learning algorithm has been widely used in many studies that predict financial distress. [11] A comparison of the Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) models showed that SVM was superior to LDA in predicting financial distress in industrial companies in Indonesia. Similarly, other studies have found that SVM is effective in handling high-frequency data, making it highly suitable for predicting financial distress using multiple variables. Furthermore, the Altman Z-Score, a widely recognized model for bankruptcy prediction, has been applied to predict bankruptcy at PT. Garuda Indonesia [12]. The effectiveness of this model can vary based on the specific characteristics of the company being analyzed, making it crucial to explore alternative models such as the Zmijewski model, which has also been used to predict the financial health of Indonesian companies [13, 14].

Predicting financial distress has become increasingly complex with the emergence of COVID-19. This is because companies worldwide are facing unprecedented new challenges that have previously impacted their financial stability. Research [15] has demonstrated the critical importance of financial ratios in predicting financial distress during this period. These significant changes require adaptive models that can analyze real-time data and respond to rapidly changing economic conditions. This capability has been demonstrated by studies using machine learning algorithms. Based on the above

research, the potential of machine learning algorithms in increasing prediction accuracy is very large. By combining traditional financial ratios with advanced machine learning techniques, researchers can develop more robust models that account for the complexity of the Indonesian economy. Future research should focus on developing these models, exploring sectoral variables, and adapting to current and future economic changes to improve financial distress prediction.

Financial distress prediction models have been developed since the 1930s, the initial method focused on financial ratios, as pioneered by Beaver [16] then developed with multivariate models such as Altman's Z-score [17]. The EWS model explores the contagious economic structure, improving the performance of crisis prediction models, and network analysis and the ML algorithm that found evidence of contagion risk [18].

A study [19] predicted US stock prices using ETE and ML, also examining their relationship with the economic crisis and Granger correlation. Another study [20] proposed a method for FS prediction that requires data sources and feature attributes. A combination of DNN and AWOA in predicting financial distress to determine model hyperparameters was carried out by [21], resulting in the AWOA-DL model outperforming other models, with an average accuracy of 95.8%.

Deep learning approaches are widely used in predicting financial distress. Challoumis et al. [22] combined fuzzy logic with a CNN deep learning model to increase prediction flexibility. This is applicable to situations where financial data is incomplete or ambiguous [23]. This method improves accuracy because it can better handle data uncertainty. In Malaysia, [24] used LSTM, GRU, and RNN with accuracies of 93%, 93%, and 90% when using multiple years of data. However, accuracy decreased when using one year of data, demonstrating the importance of historical data for maintaining prediction consistency.

Research in Indonesia shows that traditional statistical models produce accuracy levels between 34.5 and 70.2, while machine learning-based research using 30 financial variables significantly outperforms traditional models with an accuracy level of up to 90% [25]. Supporting this, Metwally et al. [25] in their study stated that the determinants of financial distress vary widely in the Middle East and North Africa (MENA) region, highlighting the importance of distinguishing between economic and financial factors [26]. While AI-based models are increasingly developing, traditional statistical models are still proven to be reliable. In Pakistan, Ashraf et al. [26] showed that the Zmijewski model remained consistent with an accuracy of 73.6% across different economic conditions [27]. Meanwhile, Pham Vo Ninh et al. [27] analyzed financial distress in 800 companies in Vietnam using a combination of accounting, market, and macroeconomic variables [28].

Research by Halteh et al. [28], states that threats to a company's financial health can arise from significant global economic events [29]. Statistical and AI-based models are playing an increasingly important role in financial and economic forecasting [30–32]. A study states that the Gaussian Process Regression algorithm is excellent at discovering complex price patterns and making accurate predictions with relatively low corrected mean squared error (RRMSE) [33]. AI models that help in a comprehensive understanding of financial variables also contribute to higher prediction accuracy and how different modeling techniques compare in predicting financial distress [34]. Ntoug et al. [34] found that the Altman Z-score model indicates lower financial risk in family firms in

Spain. The significant difference in Z-scores between family and non-family firms indicates that family-owned businesses in Spain are generally more financially stable [35]. However, studies by Pech et al. and Vukčević et al. [36, 37] have demonstrated a decline in the effectiveness of statistical models such as the Altman Z-score in predicting financial distress.

Variables such as Inventory Turnover are a measure of how quickly a business sells and replaces its inventory within a given time period. Because they indicate operational performance, this ratio is crucial in predicting financial distress [38]. Accounts Receivable Turnover is a measure of how effectively a business collects its receivables from credit sales. This factor indicates the effectiveness of cash flow management, making it highly relevant for predicting financial crises [39]. Unbalanced total asset growth—either low or sharply increasing—is often associated with financial distress. Low growth can indicate a risk of distress, while a surge in growth can indicate higher growth compared to a company that is not experiencing distress [40]. The operating income growth rate, which measures how operating income increases, is crucial for predicting financial crises because it reflects the core performance of the business.

The current ratio is crucial in predicting financial distress because it measures a company's ability to pay its short-term obligations with its current assets [41]. In predicting financial crises, the quick ratio offers a more accurate assessment of short-term liquidity by considering only assets that can be converted into cash, such as accounts receivable and cash. One important metric in predicting financial crises is the Market Value of Equity to Total Liabilities Ratio, which indicates how well a company can cover its liabilities with its equity based on market valuation. Market value of equity, also known as Market Value of Equity or MVE, indicates the stock market's valuation of a company's outstanding shares. A low MVE ratio may indicate that the market perceives the company as risky, which increases the likelihood of a financial crisis [42].

3 Material and method

3.1 Data preparation

3.1.1 Data collection

The dataset used in this study comprises annual financial data collected from 325 Indonesian companies over the period from 2013 to 2023, totalling 3575 data entries. The data is structured to include a variety of financial metrics that are indicative of a company's financial health, as well as other relevant variables. Each row in the dataset represents the financial data for a single company in a specific year. The dataset includes the following features in Table 1:

The data comes from official reports from the Indonesia Stock Exchange and the financial statements of each company. The Corruption Index data is sourced from the website <https://www.transparency.org/en/countries/indonesia>. Data processing was performed using a number of financial ratios listed in Table 1. These ratios reflect the company's internal financial condition. This study also used the company's market capitalization value. Based on these ratios and the corruption index used, the actual condition of the company is explained. The data is then tabulated according to variable type and year to ensure more precise predictions.

Table 1 Description of all features in the dataset

Symbol	Variable	Description
Y	Financial distress	A binary outcome variable indicating financial distress. A value of '1' denotes financial distress, defined as a condition where a company has negative equity for two consecutive years, while '0' indicates no financial distress
X1	Gross profit	This is the gross profit margin, which is found by dividing (sales—cost of goods sold) by sales. This indicator shows how profitable a company's main business activities are
X2	Net profit margin	This number shows the proportion of sales that turn into profit after all costs by dividing earnings after tax by sales
X3	Current ratio	This ratio shows how well a corporation can pay its short-term debts by dividing its current assets by its current liabilities
X4	Quick ratio	The quick ratio is calculated as Current Liabilities, which shows how well a company can pay its short-term debts without selling inventory.
X5	Cash ratio	This ratio shows how much of a company's assets are in cash by dividing cash by total assets. It shows how liquid the company is
X6	Receivable turnover	Shows how well a business collects its debts. A higher turnover means that the company has good policies for collecting money and giving credit
X7	Inventory turnover	Shows how fast a corporation sells and replaces its stock. Higher values mean that inventory management is working well
X8	Leverage asset	This number, which is found by dividing total liabilities by total assets, shows how much a company is in debt and how much financial risk it may face
X9	Leverage equity	This number, which is total liabilities divided by equity, gives you an idea of a company's financial structure and how much it relies on debt
X10	Debt service coverage ratio	It checks to see if a corporation can pay off its debt with its operational income by dividing its net operating income by its total debt
X11	Asset Coverage Ratio	This ratio shows how well a corporation can pay off its debt with its assets by dividing its total assets by its total long-term debt
X12	MBV (Market to Book Value Ratio)	A ratio that shows how much a company's stock is worth on the market compared to its book value. A greater ratio means that people think the stock is worth more
X13	Independent commissioner	The number of independent commissioners compared to the overall number of commissioners shows how much independent monitoring there is in a corporation
X14	Board diversity	This number shows how many women are on a company's board compared to the total number of directors
X15	Corruption index	The value is measured from the corruption level index of a country, where a higher index value has a lower level of corruption

3.1.2 SMOTE

Imbalanced data can lead to biased results. Therefore, SMOTE is necessary to balance the dataset [43]. Minority data cannot represent categorization based on specified clusters, such as distressed or non-distressed. Smote adds samples that represent the minority samples in the dataset. This addition and balancing with the Smote method makes it easier for the machine to analyze and learn two or more different classes. In a number of applications, Smote is predominantly used to detect crises, medical conditions, fraud, and specific events [43].

3.1.3 PCA

PCA reduces dimensionality by transforming the original feature set into a smaller number of uncorrelated variables called principal components (PCs) that capture the greatest possible variance. This compression helps remove redundancy arising from correlated features, can denoise the data, and often simplifies downstream models without substantial information loss.

$$C = \frac{1}{n-1} X^T X \quad (1)$$

Compute the eigen-decomposition $C = V\Lambda V^T$ where columns of V are eigenvectors and Λ diagonal matrix of eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$. The top k eigenvectors $W = [v_1, \dots, v_k]$ define the projection, and the reduced representation is

$$Z = XW \in \mathbb{R}^{n \times k} \quad (2)$$

Each eigenvalue λ_i reflects the variance explained by PC i the explained variance ratio is $\lambda_i / \sum_j \lambda_j$. Selecting k is typically based on a target cumulative variance (e.g., 90–95%) or validation performance in the downstream task.

By projecting the data onto the first few principal components, PCA reduces the dimensionality of the dataset, simplifying the model and often improving its performance by eliminating noise and redundancy. PCA is also widely used for visualizing complex data in lower dimensions (e.g., 2D or 3D plots) and as a preprocessing step in machine learning pipelines to reduce the computational burden of training algorithms. However, PCA assumes that the principal components with the highest variance are the most important, which may not always hold true for all types of data and tasks.

3.2 Traditional machine learning

3.2.1 Support vector machine

Support Vector Machine (SVM) is a powerful supervised machine learning algorithm primarily used for classification tasks but also applicable to regression. The key idea behind SVM is to find the optimal hyperplane that best separates data points of different classes in the feature space. A hyperplane in a two-dimensional space is a line dividing the space into two regions, and in higher dimensions, it's a flat subspace that separates data points [44]. In the linear case, given (x_i, y_i) with $y_i \in \{-1, +1\}$ the primal problem is.

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \text{ s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (3)$$

The resulting decision rule is $\hat{y}(x) = \text{sign}(w^T x + b)$

With feature map $\phi(\cdot)$ and kernel $K(x, x') = \phi(x)^T \phi(x')$ the decision function

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x_i, x) + b, \hat{y}(x) = \text{sign}(f(x)) \quad (4)$$

With $\{\alpha_i\}$ from the dual optimization.

3.2.2 Decision tree

A Decision Tree is a tree-like model used for decision-making and predictive modeling in machine learning, suitable for both classification and regression tasks [45]. The tree is built by recursively splitting the dataset into subsets based on the value of the input features. Each internal node of the tree represents a feature (or a decision rule), each branch represents the outcome of the rule, and each leaf node represents the final output or class.

The splitting criterion, such as Gini impurity

$$\text{Gini} = 1 - \sum_{c \in \mathcal{C}} p_c^2 \quad (5)$$

where p_i is the probability of class i in the given node, or information gain IG

$$\text{Entropy} = - \sum_{c \in \mathcal{C}} p_c \log_2 p_c, \text{IG} = \text{Entropy}(\text{parent}) - \sum_k \frac{|S_k|}{|S|} \text{Entropy}(S_k) \quad (6)$$

where $\text{entropy} = - \sum_i p_i \log_2(p_i)$, determines how the data is divided at each node. To simplify the interpretation and visualization of predictive models, it's best to use decision trees. However, decision trees can produce biased predictions due to the complexity and noise of the data used. Therefore, to address this issue, decision trees need to be combined with ensemble learning for optimal prediction results. There are two models derived from decision trees based on ensemble learning random forest and gradient boosting.

Random Forest is a method that builds multiple decision trees during training and outputs the mode of the classes (for classification) or the mean prediction (for regression) of the individual trees. By averaging the results, Random Forest reduces the variance of the model, making it more robust and less prone to overfitting compared to a single decision tree [46]. Random Forests are highly accurate, robust to noise, and can handle large datasets with high dimensionality. They also provide feature importance metrics, indicating which features contribute most to the model's predictions. However, Random Forests can be computationally intensive due to the large number of trees, and the model's interpretability is reduced compared to a single decision tree. On the other hand, gradient boosting builds models sequentially, each new model trying to correct the errors made by the previous ones. In Gradient Boosting, decision trees are typically used as the base learners. However, unlike Random Forests, where trees are built independently, in Gradient Boosting, each tree is trained to fix the errors of the previous trees [47]. The "gradient" part refers to the use of gradient descent to minimize the loss function, which measures how well the model fits the training data. Gradient Boosting is powerful and can achieve high predictive accuracy, but it requires careful tuning of hyperparameters to prevent overfitting and manage the computational cost.

3.2.3 Random forest

Random Forest is an ensemble learning method for classification and regression that constructs a collection of decision trees and aggregates their predictions to improve generalization performance [48]. Each tree is trained on a bootstrap sample of the training data, while at every split, a random subset of features is considered (feature subsampling). This twofold randomisation over observations and features reduces variance relative to a single tree and mitigates overfitting. In the context of classification, the forest's prediction is typically determined by the majority vote across trees. In contrast, for regression, the forest's prediction is the average of all individual predictions.

Formally, let $\{T_b(\cdot)\}_{b=1}^B$ denote decision trees trained on bootstrap samples with feature subsampling. For an input x , the Random Forest classifier returns

$$\hat{y}(x) = \arg \max_{c \in C} \sum_{b=1}^B 1 \{T_b(x) = c\} \quad (7)$$

For regression, the prediction is

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (8)$$

Key hyperparameters include the number of trees (n_estimators), the number of candidate features per split (max_features), maximum depth (max_depth), and split/leaf size (min_samples_split, min_samples_leaf). Class imbalance can be addressed via class weights or train-split resampling (e.g., SMOTE), applied only within the training data, no resampling is performed on validation or test sets.

Random Forests are robust to noisy features and outliers and naturally capture nonlinearities and higher-order interactions. They handle mixed-scale variables without feature scaling and offer feature importance diagnostics (e.g., permutation importance).

3.2.4 K-nearest neighbor

K-Nearest Neighbors (KNN) is a simple, non-parametric, and instance-based learning algorithm that can be used for both classification and regression tasks. Unlike many other algorithms that rely on a predefined model, KNN makes predictions based directly on the dataset. When a new data point is introduced, KNN looks at the 'k' closest data points in the training dataset (measured by a distance metric such as Euclidean distance) and assigns the most common label (in classification) or the average value (in regression) of these neighbors to the new point [49]. Predictions are made directly from the training set x find the indices for $\mathcal{N}_k(x)$ of the k nearest training samples under a chosen distance. The classification rule is

$$\hat{y}(x) = \operatorname{argmax}_{c \in C} \sum_{i \in \mathcal{N}_k(x)} 1 \{y_i = c\} \quad (9)$$

KNN is intuitive and easy to implement, making it popular for small datasets where interpretability and simplicity are important. However, KNN has several drawbacks it can be computationally expensive since it requires calculating the distance to all training points for each prediction, and it is sensitive to the choice of 'k' and the distance metric. Additionally, KNN doesn't perform well with high-dimensional data or imbalanced datasets, as the majority class can dominate the nearest neighbors.

3.3 Artificial intelligence

The use of Artificial Intelligence (AI) is unavoidable in everyday life. AI mimics the neural networks of the human brain, consisting of layers of interconnected neurons, allowing the network to generate decisions. Artificial Neural Networks (ANNs) are used to make decisions based on data streams within complex neural networks. Therefore, these neural networks must first analyze and classify them before making a decision.

Decision-making in ANNs does not utilize the linearity of neural networks, allowing them to handle large amounts of diverse data patterns. Datasets are used to continuously train the network, allowing it to adapt to existing data patterns and improve its

prediction results [50]. As the network grows, new predictive models emerge, allowing the model to adapt to complex and abstract data. Deep learning from neural networks encompasses not only numbers and letters, but also images and sounds, which can be learned by the neural network as data. The ability to automatically read complex raw data is tailored to the user's needs.

4 Results

4.1 Data processing

Data were collected from the financial statements of 325 firms between 2013 and 2023, yielding 3,575 observations and have cleaned too 3,573 row data. All the Y and X1–X15 variables in Table 1 were computed from these statements. Variables X1–X14 originate from the companies' financial statements, while X15 is the Indonesian Corruption Index. A descriptive summary of all features is provided in Table 2.

In preparation for developing the model, we performed basic preprocessing (type checks, handling of missing values and column normalisation) and inspected outliers using box-and-whisker plots. We identified a small number of extreme outliers in X1 and X2 that were located far from the majority of the data. As these points represent a very small proportion of the full sample, we retained them to avoid distorting the original distributions. For the remaining variables, only minor local outliers are present, and the distributions are reasonably centred. To contextualise the evaluation within the properties of our data, we visualised the class imbalance and demonstrated the effect of applying SMOTE exclusively to the training set (the validation and test splits remained unaffected). These diagnostics explain why naïve models can achieve high accuracy yet underperform on the minority class (low F1 score), and motivate our decision to rebalance within stratified folds while preserving leakage-free validation and an untouched external test set.

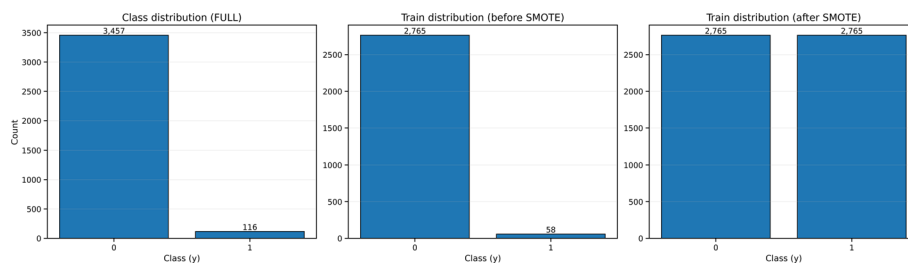
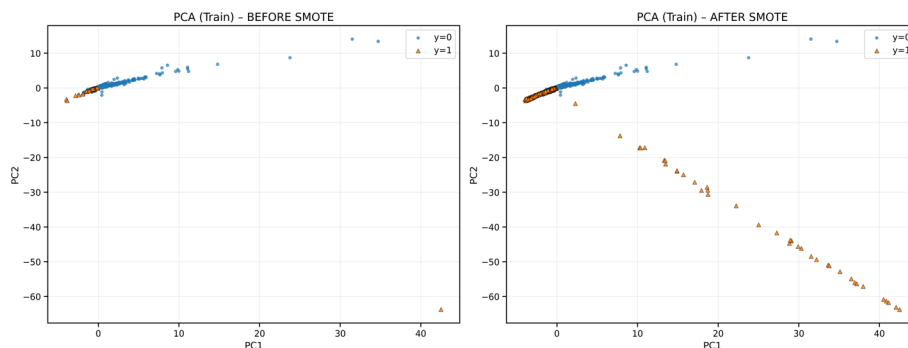
The findings in Table 3 reveal a clear trade-off between overall accuracy and minority-class performance, a common challenge in imbalanced datasets. While all models achieved high accuracy (95–99%), their F1-scores, the primary selection criterion for this task, varied significantly. The Decision Tree model emerged as the top performer,

Table 2 Statistical description of all features in the dataset

Ft	Mean	Std	Min	Max	Var	Skew	Kurt
Y	0,032	0,177	0,000	1,000	0,031	5,278	28.873
X1	−12,715	2187,557	−130,563,180	100,000	4,785,404.300	−59.543	3557.098
X2	−46,093	2572,515	−149,188,860	17,933,000	6,617,832,737	−54,613	3170.503
X3	2,484	7,660	0,010	247,127	58,676	21,078	546.272
X4	1,638	6,283	0,000	246,835	39,477	28,030	929.481
X5	0,943	5,881	−0,226	246,724	34,589	30,210	1085.503
X6	14,209	40,911	−15,694	983,170	1673,746	14,039	268.845
X7	23,043	105,845	−26,974	2435,280	11,203,220	14,523	264.659
X8	31,682	41,663	0,000	987,255	1735,767	8,518	143.870
X9	76,223	250,376	−3493,000	5018,982	62,688,364	2,646	101.406
X10	0,266	2,045	−13,837	59,866	4,181	19,315	453.385
X11	17,103	255,800	−1649,641	12,150,411	65,433,767	40,375	1769.055
X12	5,047	32,744	−95,867	648,531	1072,140	12,669	185.471
X13	0,399	0,155	0,000	3,000	0,024	2,489	39.078
X14	0,137	0,196	0,000	5,000	0,039	5,135	108.275
X15	36,092	2,235	32,000	40,000	4,993	−0,164	2.189

Table 3 Best hyperparameters obtained for each model type

Model	Hyperparameter	Accuracy	F1-score
SVM	{'clf__C': 10, 'clf__gamma': 'scale', 'clf__kernel': 'linear'}	0.9860	0.8214
Decision tree	{'clf__criterion': 'entropy', 'clf__max_depth': 5, 'clf__min_samples_leaf': 1, 'clf__min_samples_split': 2}	0.9930	0.8980
Random forest	{'clf__max_depth': 10, 'clf__max_features': 'sqrt', 'clf__min_samples_leaf': 2, 'clf__min_samples_split': 2, 'clf__n_estimators': 100}	0.9888	0.8462
KNN	{'clf__n_neighbors': 3, 'clf__p': 1, 'clf__weights': 'uniform'}	0.9510	0.5333
Gradient Boosting	{'clf__learning_rate': 0.1, 'clf__max_depth': 2, 'clf__min_samples_leaf': 5, 'clf__n_estimators': 50, 'clf__subsample': 1.0}	0.9902	0.8627

**Fig. 1** Class imbalance and the effect of SMOTE (training only)**Fig. 2** PCA view of training data before vs. after SMOTE

achieving the highest F1-score (0.8980) while also maintaining excellent accuracy (0.9930). This was followed by Gradient Boosting (F1: 0.8627) and Random Forest (F1: 0.8462). The SVM model delivered reasonable performance (F1: 0.8214), but the KNN model (F1: 0.5333) was notably ineffective at identifying the minority class. The significance of this result is that for this particular problem, models capable of robust, non-linear class-boundary shaping (specifically Decision Trees and Boosting) are demonstrably more effective. They successfully identify the difficult-to-find minority labels without sacrificing overall model accuracy.

Figure 1 shows the class imbalance from the full dataset in Class Distribution the full dataset is heavily skewed towards the non-distressed class ($Y=0$). In the Train distribution (before SMOTE) shows the training split before SMOTE, which inherits this skew. In Train distribution the training split after SMOTE balancing is balanced ($Y=0 \approx Y=1$) while the validation and test splits remain untouched, This balancing is only used for model fitting all evaluations are performed on clean splits to avoid leakage.

Figure 2 shows that the minority samples ($Y=1$) are sparse in the projected space prior to SMOTE. Systematic neighbours define the minority manifold, sharpening the

decision boundary after SMOTE. Both panels use the same PCA axes (based on the original training set) to enable direct comparison of changes.

We use an end-to-end evaluation pipeline with stratified folds, applying SMOTE only to the training splits. Starting from the full dataset, we first create an external 20% test set that remains completely clean no SMOTE or rescaling is applied to it. The remaining 80% training pool is then partitioned into 5 stratified folds. For each of these folds, we apply SMOTE exclusively to the training split to balance the minority class, while leaving the corresponding validation split untouched. Models are trained on the SMOTE-augmented training split and evaluated on the clean validation split for that fold. Once all folds have been completed, a final model is trained on the entire 80% training pool (with SMOTE) and performance is reported on the untouched 20% test set. This design preserves the temporal and label integrity of the test data, prevents data leakage from SMOTE and produces reliable metrics (Accuracy, Precision/Recall/F1, ROC-AUC, MCC and Kappa), which we later aggregate and compare across models (Figs. 3 and 4).

4.2 Feature analysis

The analysis of the relationships between variables using Pearson correlation (Fig. 5) and the p value map (Fig. 6) provides a marginal (pairwise) overview of the association of each feature with financial distress. It has been demonstrated that a number of pairs demonstrate low p values, thus enabling the statistical detection of a simple linear relationship. However, the magnitude of the correlation coefficients ($|r|$) for many pairs is relatively small, indicating that their substantive effect is weak from a bivariate perspective. Consequently, the interpretation of the results must consider two dimensions simultaneously: The primary focus of this study is twofold. Firstly, statistical significance (p value) and, secondly, effect size (the value of $|r|$). It is also acknowledged that inter-factor correlations (collinearity) are apparent in the correlation map.

In Fig. 6. The p -value heatmap for two-sided Pearson tests. This map complements the correlation matrix by showing statistical significance. Dark blue cells indicate small p -values (e.g., $<.001$), providing strong evidence against H_0 (i.e., the correlation is statistically significant). Red and light-colored cells indicate large p -values, suggesting

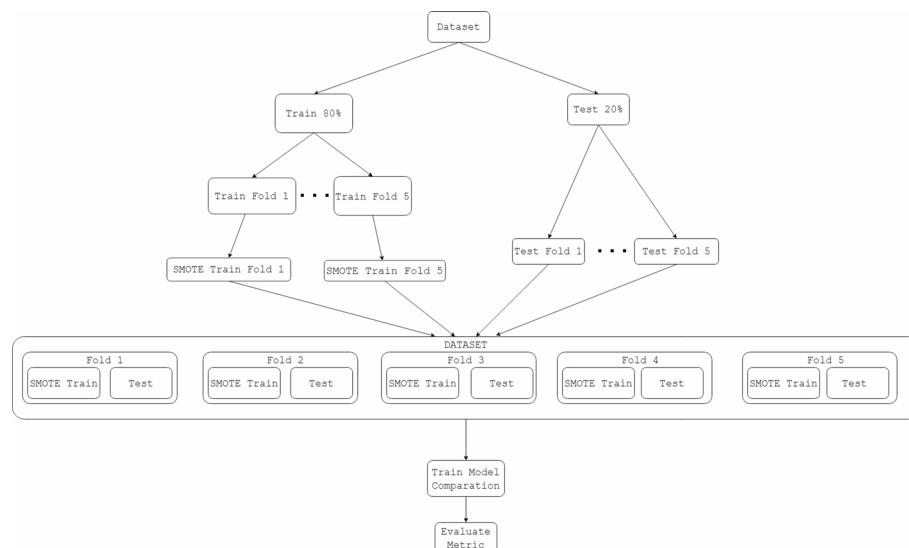


Fig. 3 Training & test flow model

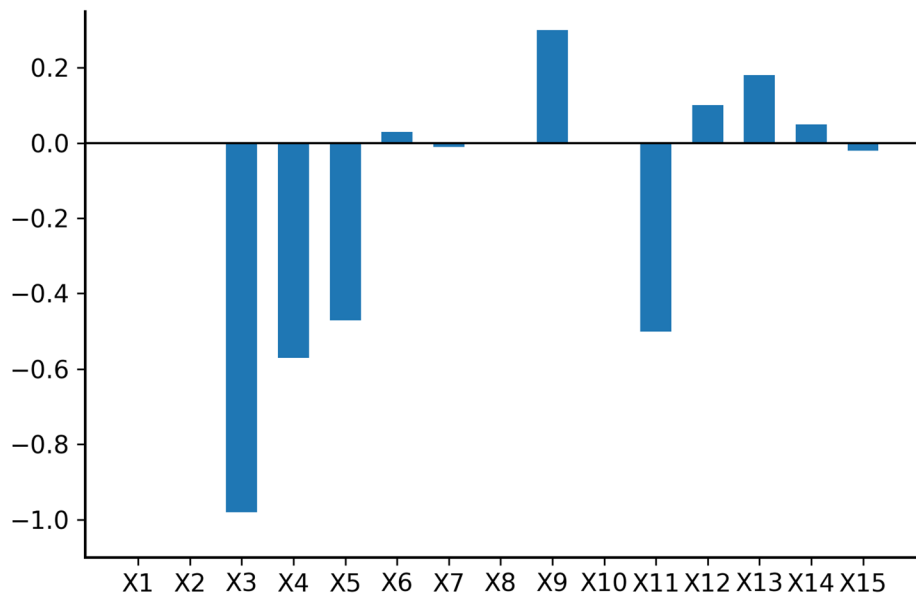


Fig. 4 Weight of each variable in logistic regression analysis to the data

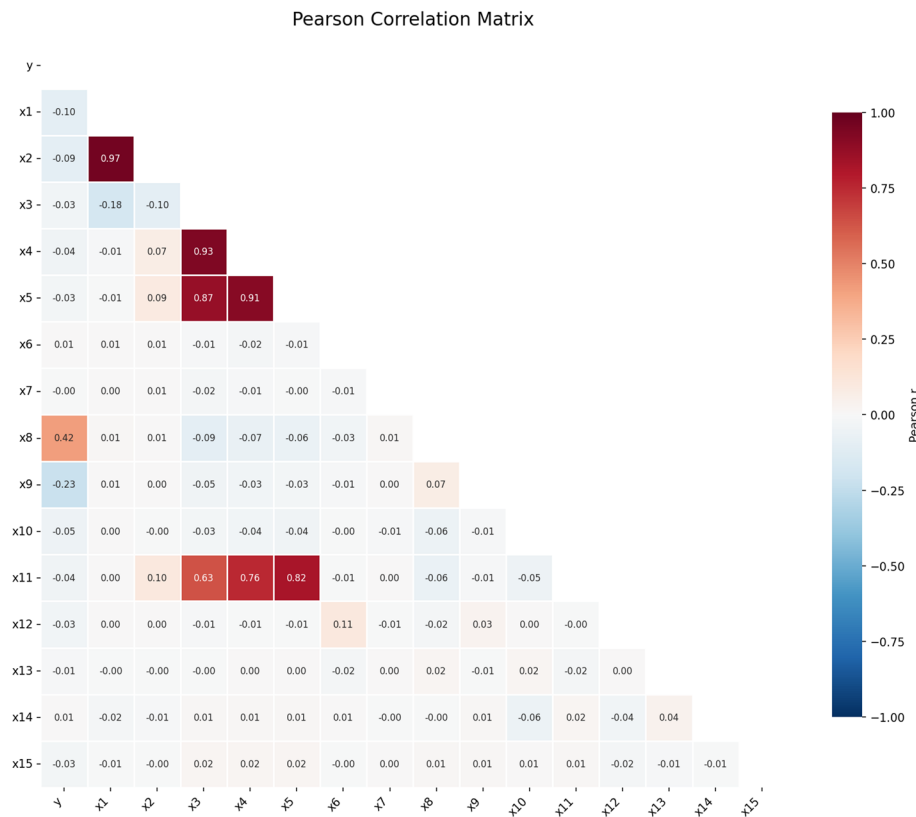


Fig. 5 Matrix of Pearson's correlation coefficient of the data

no evidence of a linear association. It is important to note that statistical significance only reflects evidence of linear dependence practical relevance should be judged by the magnitude of the correlations in Fig. 5. As many pairwise tests were performed, false discovery rate control (e.g. Benjamini–Hochberg) could be applied if strict multiplicity

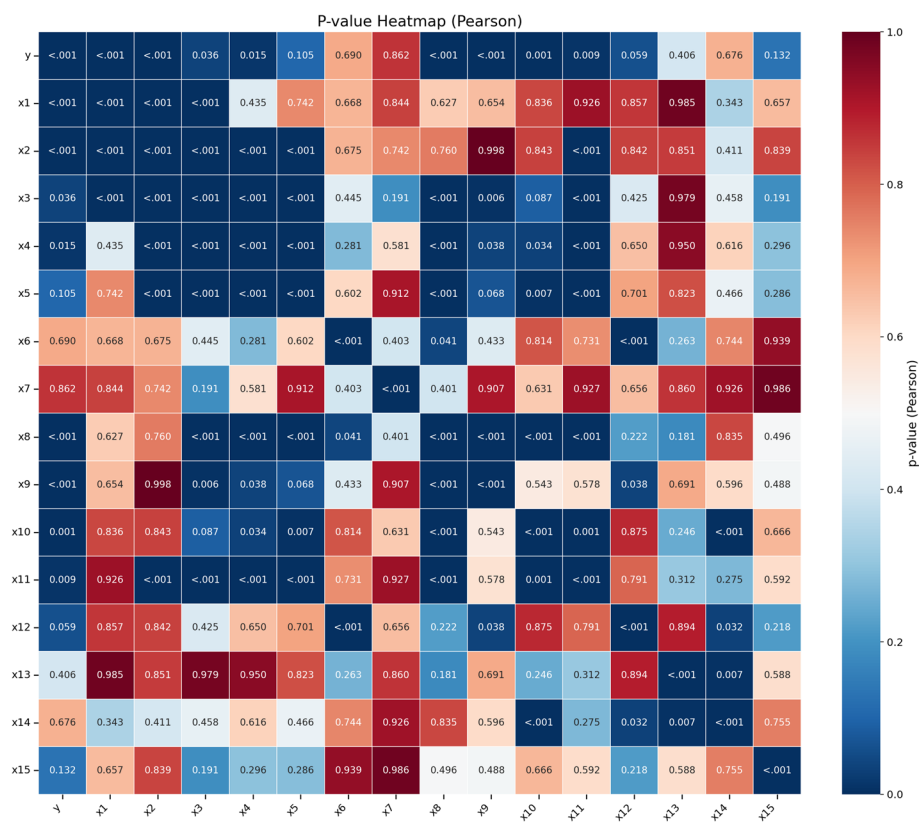


Fig. 6 Matrix of Pearson's correlation coefficient of the data its *p* values

control is desired. this heatmap visualizes the statistical significance of the correlations presented in Fig. 5. It helps separate relationships that are statistically meaningful from those that could be due to random chance. Dark blue cells indicate a *p*-value less than the 5% significance level ($p < 0.05$). This means we have strong evidence that a linear relationship exists. Light/red cells ($p > 0.05$) indicate no evidence of a linear relationship. The first row and column are the most important, showing which predictors are statistically linked to the target variable Y. Significant Predictors: X1, X2, X3, X4, X8, X9, X10, and X11 all have dark blue cells ($p < 0.05$), indicating their linear relationships with Y are statistically significant. Non-Significant Predictors: X5, X6, X7, X12, X13, X14, and X15 have light-colored cells ($p > 0.05$). This suggests their weak correlations with Y (seen in Fig. 5) are not statistically significant and likely uninformative. This map also confirms the relationships between the predictors. The large, dark blue blocks (e.g., between pairs in the X1–X5 group) show that the strong correlations (multicollinearity) observed in Fig. 5 are also highly statistically significant.in contrast, pairs with very large p-values (e.g., X2 and X9, $p = 0.998$) confirm a significant lack of linear association, reinforcing that they are independent of each other.Overall, this p-value map validates our findings from the correlation matrix, allowing us to confidently identify which variables (X1, X2, X8, etc.) are statistically linked to the target, while also confirming which predictors (X1–X5 group) are redundantly correlated with each other.

4.3 Key predictors of financial distress: liquidity and asset management

Both Figs. 5 and 6 highlight the central role of liquidity and asset management in predicting financial distress.

4.3.1 Liquidity ratios (X3: current ratio, X4: quick ratio, X5: cash ratio)

The Current Ratio (X3), Quick Ratio (X4), and Cash Ratio (X5) exhibit a weak negative correlation with distress (approximately $Y-X3 \approx -0.03$, $Y-X4 \approx -0.04$, $Y-X5 \approx -0.03$). In Fig. 6, these pairs generally have low p values (statistically detectable for a large sample size), but their effect is small in a marginal linear sense. In conclusion, higher liquidity tends to be associated with a decrease in the incidence of distress, although the magnitude of this association is small when viewed on a per-variable-pair basis.

4.3.2 Asset coverage ratio (X11)

The correlation between $Y-X11$ is observed to be small and negative (≈ -0.04) and is accompanied by a low p value. Thus, this simple linear relationship is detected despite its moderate-to-weak effect. From an economic perspective, this outcome aligns with the notion that the capacity of assets relative to long-term liabilities functions as a balance sheet buffer, thereby reducing the probability of financial distress.

4.4 The role of leverage: a double-edged sword

Leverage encapsulates a firm's reliance on debt financing and, as shown in the figures, bears asymmetrical implications depending on its construction.

4.4.1 Leverage asset side and equity side (X8 & X9)

The Leverage Asset variable (X8) exhibited a moderate positive correlation with distress (approximately $+0.42$) with a low p value, suggesting a systematic pattern where asset based leverage is associated with an increased risk of distress from a marginal viewpoint. In contrast, Equity Leverage (X9) exhibited a negative correlation (approximately -0.23) with a low p value. This finding suggests that a larger equity buffer is associated with a reduction in distress risk. In summary, asset gearing increases vulnerability, whereas equity capacity provides a protective effect from a linear bivariate perspective.

4.4.2 Profitability: gross profit (X1) and net profit margin (X2)

The profitability of the enterprise is influenced by a number of factors, including X1 and X2. Gross Profit (X1) and Net Profit Margin (X2) demonstrate weak negative correlations with distress (approximately -0.10 and -0.09 , respectively). Despite the low p values observed for these pairs (indicative of a detectable effect), their effect sizes are minimal. Consequently, from a marginal perspective, profitability is not a robust single indicator of solvency risk. This finding is consistent with the notion that margins can be sustained even as balance sheet strength declines.

4.4.3 Market perception: board of directors diversity (X14) corruption index (X15)

The correlation between board diversity (X14) and distress ($\approx +0.01$) is negligible, and the p value is not significant. Concurrently, the Corruption Index (X15) exhibited a negligible negative correlation (approximately -0.03) with a p value that did not reach statistical significance. Consequently, from a marginal linear standpoint, neither shows a

substantive association with distress. Should any influence be demonstrated, it is likely to be indirect, conditional on other variables, or non-linear, and therefore not captured by a simple Pearson correlation.

4.4.4 Efficiency measures: limited predictive power

Accounts receivable turnover (X6) and inventory turnover (X7) have a very low correlation in predicting financial distress, as seen in Figs. 4 and 5. Efficient management of accounts receivable and inventory will help reduce a company's operating costs, resulting in faster cash flow. Stable cash flow will reduce the company's operating expenses, eliminating the need to seek new financing through new share issuances or new debt.

Figure 5 Shows strongest multicollinearity is observed between X1-X2 with value approximately 0.93, and X4-X5 also exhibiting a high degree of multicollinearity approximately 0.91. These large positive correlations suggest that these features contain overlapping information and are potentially redundant. With respect to the target variable, Y exhibits its strongest linear association with X8, which has a moderately negative value of approximately -0.42, and a weaker negative association with X9, with a value approximately -0.23. Correlations, indicating little linear association (e.g. X6, X13, X14 and most little contribution X7). In contrast, the highly correlated groups (e.g. feature selection, regularisation or dimensionality reduction, such as principal component analysis (PCA)) prior to model fitting. The heatmap's colour scale reflects the strength and sign of the correlations deep red for strong positive correlations and deep blue for strong negative correlations.

To complement the correlation matrix, we performed two-sided Pearson tests on every pair of variables ($H_0: \rho = 0$) and visualised the resulting p values as a heatmap (see Figure X). Dark blue cells indicate small p values (strong evidence against H_0), while red and white cells indicate large p values (no evidence of a linear association). The values in the cells are rounded p values. Please note that statistical significance only reflects evidence of linear dependence effect size and practical relevance should be considered alongside the magnitude of the correlations. As many pairwise tests are performed, false discovery rate control (e.g. Benjamini–Hochberg) can be applied if strict multiplicity control is desired.

Figure 6 p value heatmap reveals several significant associations with the target variable Y at the 5% level of significance, most notably with variables X1, X2, X3, X4, X8, X9, X10 and X11 (dark cells in the first row and column). By contrast, X5, X6, X7, X12, X13, X14 and X15 exhibit non-significant p values with respect to Y, indicating weak linear relationships in this sample. Among the predictors, there are also blocks of very small p values (e.g. between some pairs in the X1–X5 group), which are consistent with the multicollinearity observed in the correlation matrix. In contrast, other pairs exhibit large p values (e.g. near-independence, such as between X2 and X9), which indicates little evidence of linear association. Overall, the p value map corroborates the earlier correlation findings and helps to identify variables that are statistically linked to the target variable, as opposed to those that are likely to be uninformative in a strictly linear sense.

4.5 Model construction and training

The conventional learners (Support Vector Machine (SVM), Decision Tree, Random Forest, K-Nearest Neighbour (KNN) and Gradient Boosting) were tuned using grid

search with fivefold stratified cross-validation, with SMOTE applied exclusively within the training folds to prevent leakage. As the use case prioritises the identification of financial distress ($Y=1$), both accuracy and the F1-score for the minority class are reported. The models are primarily ranked by the F1-score, with accuracy serving as a secondary verification measure.

The findings indicate a discernible trade-off across the models examined. The Decision Tree model demonstrates the highest F1-score (0.8980) and strong Accuracy (0.9930), thus establishing itself as the top performer in accordance with the established selection criterion. Gradient Boosting follows ($F1=0.8627$ $Acc=0.9902$), while Random Forest delivers competitive Accuracy but a lower F1 (0.8462). The SVM model demonstrated reasonable performance, with an F1 score of 0.8214, while the KNN model exhibited inferior performance for the minority class, with an F1 score of 0.5333. The findings of this study demonstrate that models characterised by robust class-boundary shaping (Decision Tree/Boosting) are more effective in accurately identifying minority labels without compromising overall accuracy.

Even though it already gives a very good result, we proceed to apply the same experimentation to the deep learning models. We expect it to give even higher results. In order to investigate the impact of network capacity and nonlinearity on the detection of minority class data, a series of experiments were conducted. In these experiments, artificial neural networks (ANNs) with single hidden layers were evaluated across a range of neuron counts (20, 50, 100, 200) and activation functions (ReLU, Tanh, Sigmoid). The accuracy and F1-score for the positive class ($Y=1$) are reported, denoted here as $F1(Y=1)$, since the primary objective is to correctly identify instances of financial distress. The training of models is conducted within a leakage-safe setup, employing SMOTE exclusively on training folds. The performance of these models is subsequently measured on a clean hold-out test set.

The results obtained from the study in Table 4.

The primary objective, maximizing the F1-Score for the positive class ($F1(Y=1)$), was evaluated alongside overall accuracy. The optimal balance between minority-class detection and overall accuracy was achieved by the 50-Sigmoid configuration (50 neurons, Sigmoid function). This model yielded the highest stable F1-Score at 0.9500 and tied for the highest accuracy at 0.9920. This suggests a moderately compact network, in conjunction with a smooth, bounded nonlinearity (Sigmoid), is exceptionally effective at capturing the minority decision boundary without overfitting. This have several findings

Table 4 Accuracy and F1-score of models with different neurons and activation function

Neurons	Activation	Accuracy	F1-score
20	ReLU	0.9920	0.9492
	Tanh	0.9907	0.9412
	Sigmoid	0.9907	0.9421
50	ReLU	0.9867	0.9138
	Tanh	0.9893	0.9322
	Sigmoid	0.9920	0.9500
100	ReLU	0.9853	0.9060
	Tanh	0.9827	0.8870
	Sigmoid	0.9907	0.9412
200	ReLU	0.9827	0.8850
	Tanh	0.9827	0.9906
	Sigmoid	0.9893	0.9322

key, Diminishing Returns: For most activations, performance peaked early. Both ReLU and Sigmoid models showed diminishing returns, with performance gains becoming non-systematic or degrading as network width increased beyond 50–100 neurons. This implies that simply adding more neurons is an ineffective strategy here. The ReLU Alternative: The 20-ReLU model (20 neurons, ReLU) provided a robust alternative, delivering performance nearly identical to the 50-Sigmoid winner (F1: 0.9492, Acc: 0.9920). This highlights that a simple, non-saturating activation can also be highly effective in a very compact network. A High-F1 Anomaly: A notable exception was the 200-Tanh model, which achieved the absolute highest F1-Score (0.9906). However, this high F1 came at the cost of lower accuracy (0.9827) and required a much larger network. This result appears to be a high-variance outlier rather than a stable, generalizable trend. If the primary objective is the most robust and balanced detection of financial distress, the 50-Sigmoid configuration is the optimal choice. If simplicity and high accuracy are preferred, the 20-ReLU model is a virtually identical and highly robust alternative. which tested various network capacities and activation functions, indicate a clear and complex trade-off.

We want to further improve these results by considering other hyperparameters outside the architecture. In the context of deep learning model, there are hyperparameters that are not related to the internal architecture of the models but related to the training setup of the model. These hyperparameters are number of epochs and batch size of datapoints put to the model for each iteration. Utilising the architectural comparison presented in Table 4, the artificial neural network (ANN) is configured to the optimal activation width configuration, which is determined to be a single hidden layer comprising 50 neurons, in conjunction with a sigmoid activation function. The subsequent study of training dynamics is achieved by varying two optimisation hyperparameters epochs and batch size within a single leakage-safe protocol (SMOTE applied exclusively to training folds validation/test remain unaltered). The models are trained for epochs ranging from 20 to 200 and batch sizes ranging from 2 to 512. The primary measure of effectiveness in detecting financial distress is reported as accuracy and the F1-score for the positive class ($Y = 1$), denoted $F1(Y = 1)$.

As demonstrated in Table 5, show a clear and predictable relationship between epochs, batch size, and model performance. Using the optimal 50-Sigmoid architecture, the analysis reveals two key trends The Sweet Spot for Batch Size, there is a distinct sweet spot for model performance. Moderate batch sizes (8–32) consistently produce the strongest F1-Scores. Conversely, very small batches (e.g. 2) show slightly lower performance, while very large batches (128–512) clearly underperform, with the largest batch sizes causing a significant drop in F1-score. The Epoch Plateau: Performance (F1-score) improves with longer training, moving from an F1 of ~ 0.90 at 20 epochs to a higher, stable level. The results reach a plateau at approximately 100–200 epochs, where further training provides no additional benefit. By combining these trends, we find that the peak performance is achieved at an F1-Score of 0.9500 and an Accuracy of 0.9920. The findings show that this optimal, balanced result is reached by two identical-performing configurations: Epochs: 100, Batch Size: 8 Epochs: 200, Batch Size: 32. This suggests that once the 50-neuron Sigmoid architecture is set, training it for 100–200 epochs using a moderate batch size (8 or 32) provides the optimal and most stable balance between minority-class detection and overall accuracy..

Table 5 Accuracy and F1-score of models with different epochs and batch size

Epochs	Batch size	Accuracy	F1-score
20	2	0.9893	0.9333
	8	0.9853	0.9106
	32	0.9840	0.9032
	128	0.9720	0.8421
	512	0.9320	0.6623
50	2	0.9867	0.9194
	8	0.9907	0.9421
	32	0.9867	0.9194
	128	0.9840	0.9032
	512	0.9653	0.8116
100	2	0.9880	0.9231
	8	0.9920	0.9500
	32	0.9907	0.9421
	128	0.9840	0.9032
	512	0.9773	0.8682
200	2	0.9880	0.9231
	8	0.9893	0.9322
	32	0.9920	0.9500
	128	0.9907	0.9421
	512	0.9840	0.9032

4.6 Comparing model prediction analysis

This part provides an in-depth analysis of the performance of various machine learning models in predicting financial distress among Indonesian companies, integrating insights from the results and contextualizing them within established domain knowledge of financial distress prediction.

4.6.1 Performance of traditional machine learning models

The benchmarking process involved five conventional classifiers Support Vector Machine (SVM), Decision Tree, Random Forest, k-Nearest Neighbours (KNN), and Gradient Boosting. These were evaluated under a leakage-safe protocol, which involved the application of SMOTE only within the training folds, while the validation set and the external test set were left unaltered. The present study aims to emphasise the detection of financial distress by reporting overall accuracy alongside $F1(Y = 1)$.

4.6.2 Support vector machine (SVM)

The SVM with a linear kernel attains an accuracy of 0.9860 and an F1 score of 0.8214 when predicting the outcome Y as 1. The findings suggest that a linear decision boundary successfully captures a significant proportion of the signal while maintaining a high level of overall accuracy. It is posited that the incorporation of additional nonlinearity (e.g., radial basis functions or polynomial kernels) or calibrated decision thresholds could further enhance the retrieval of minority classes. However, it is imperative to exercise caution with regard to regularisation in order to circumvent the risk of overfitting.

4.6.3 Decision tree

The Decision Tree model attains an accuracy of 0.9930, with the highest $F1(Y = 1)$ score of 0.8980, thus demonstrating superior performance in comparison to the conventional models. This finding indicates that a constrained tree (depth=5, entropy criterion)

effectively captures salient interactions in the ratios without excessive variance. Although trees are interpretable and align with domain reasoning, they can be sensitive to small data perturbations therefore, stability checks and complementary models are advisable.

4.6.4 Random forest

The Random Forest model demonstrated an accuracy of 0.9888 and an F1 score of 0.8462 for the $Y=1$ classification. The ensemble demonstrates an enhancement in robustness when compared with a single tree however, under the current hyperparameters, it exhibits a decline in performance relative to the Decision Tree and Gradient Boosting on $F1(Y=1)$. The incorporation of additional tuning mechanisms, such as the inclusion of more estimators, the utilisation of class-balanced criteria, or the implementation of threshold adjustment, has the potential to further reduce this disparity. The feature importance from the forest remains valuable for diagnostic insight.

4.6.5 K-nearest neighbors (KNN)

The KNN records accuracy was found to be 0.9510, and the $F1(Y=1)$ was 0.5333. This indicates that there is limited effectiveness for minority-class detection in this feature space. Distance-based methods are sensitive to scaling, local density, and class imbalance even with SMOTE in training folds, KNN can struggle when minority patterns are sparse or anisotropic. In this context, KNN is better positioned as a baseline or complementary reference.

4.6.6 Gradient boosting

The Gradient Boosting model demonstrates a noteworthy level of accuracy, with a score of 0.9902, and an F1 score of 0.8627 when predicting the outcome as $Y=1$. This places it among the top performers in this field of study. The stage-wise additive modelling appears to be an effective method for capturing nonlinear relationships in liquidity, leverage and profitability indicators. This is achieved while maintaining control of overfitting via shallow trees and learning-rate shrinkage.

Overall, Decision Tree provides the strongest $F1(Y=1)$ with excellent Accuracy, followed by Gradient Boosting and Random Forest. SVM performs competitively under a linear margin but may benefit from nonlinear kernels and threshold calibration. KNN underperforms on the minority class in this setting. For practice, we recommend selecting models primarily by $F1(Y=1)$ (with a minimum recall target), then validating with macro-F1, MCC, and calibrated decision thresholds.

4.7 Performance of deep learning models

Deep Learning can generate a number of models for users based on financial report data, both internal company data and macroeconomic data. However, to use Deep Learning, users must first understand what models to create and what data to use.

4.7.1 Effect of epochs and batch sizes

The experimental findings suggest that a combination of 100 epochs with a batch size of 8 and 200 epochs with a batch size of 32 attains the maximum test accuracy of 0.9920, while concurrently exhibiting robust minority-class performance. In general, moderate batch sizes (8–32) provide a favourable balance between gradient noise and convergence

stability, leading to superior $F1(Y=1)$ relative to very small (e.g., 2) or very large (≥ 128) batches. The use of smaller batches can introduce higher stochasticity, which has the potential to hinder stable optimisation. Conversely, overly large batches may reduce the beneficial regularisation effect of stochastic gradients, thereby slowing progress towards minority-sensitive optima.

From a domain perspective, financial distress is shaped by a combination of internal firm characteristics (e.g. profitability, liquidity, leverage, governance) and external conditions (e.g. market dynamics, event shocks, macroeconomic factors). Consequently, the learning problem is enhanced by sufficiently large and diverse datasets and by training regimes that allow models to generalise across heterogeneous regimes. The observed performance plateau around 100–200 epochs with moderate batches suggests an efficient training regime that captures salient patterns without overfitting, aligning with the study's goal of reliable minority-class detection.

4.7.2 Effect of neurons and activation functions

The 50-neuron Sigmoid architecture was identified as the most effective configuration for this study, demonstrating robust performance for both distress and non-distress classes. This outcome is consistent with the Sigmoid function's suitability for binary classification, where its probabilistic output facilitates thresholding and operating-point tuning. Concurrently, ReLU and Tanh variants attained competitive outcomes, underscoring the notion that activation selection should be aligned with data characteristics and regularization requirements rather than regarded as universally optimal.

From a financial perspective, these activations facilitate the capture of both linear and non-linear structures. To illustrate this, consider the approximately proportional effects of leverage ratios (near-linear regions) and threshold-like behaviours in liquidity measures (non-linear regions). The 50-unit width appears to strike an optimal balance between expressive capacity and generalization, thus avoiding the underfitting risk of narrower models and the overfitting tendency sometimes observed with substantially wider layers. The findings provide robust support for the selection of activation–width pairs through empirical validation. In particular, the results indicate that the Sigmoid-50 activation function is a strong default choice for binary financial-distress prediction under the current data regime.

4.7.3 General performance analysis

In order to assess external validity under realistic class proportions, all models were evaluated on the original imbalanced test set (no SMOTE applied to test data). A comprehensive set of Test metrics is presented in Table 6, including the confusion matrix, sensitivity and recall for both classes, specificity, precision for both classes, $F1(Y=1)$, balanced accuracy, mutual information (MI), Cohen's kappa, the area under the curve (AUC) of the receiver operating characteristic (ROC), prevalence, and the nearest-neighbour index (NIR). Given a prevalence of 0.0773 the test-set prevalence due to your custom split (half of minority to test), while 0.032 is the full-dataset prevalence. State both explicitly. and an NIR of 0.9227, the overall accuracy of the model is inadequate as an indicator of minority-class performance. Consequently, the focus is directed towards $F1(Y=1)$, Recall($Y=1$), Balanced Accuracy, and MCC.

Table 6 Results of evaluation Test metrics of all models

Model		Logistic regression	SVM	Decision tree	Random forest	KNN	Gradient boosting	Artificial neural network
Confusion matrix	TN FP	679 13	684 8	689 3	689 3	668 24	689 3	687 5
	FN TP	2 56	1 57	4 54	3 55	12 46	3 55	1 57
Sensitivity (Recall)	Class 0	0.9812	0.9884	0.9957	0.9957	0.9653	0.9957	0.9928
	Class 1	0.9655	0.9828	0.9310	0.9483	0.7931	0.9483	0.9828
Specificity		0.9812	0.9884	0.9957	0.9957	0.9653	0.9957	0.9884
Precision	Class 0	0.9971	0.9985	0.9942	0.9957	0.9824	0.9957	0.9985
	Class 1	0.8116	0.8769	0.9474	0.9483	0.6571	0.9483	0.9194
F1 score	Class 0	0.9891	0.9935	0.9949	0.9957	0.9738	0.9957	0.9957
	Class 1	0.8819	0.9268	0.9391	0.9483	0.7188	0.9483	0.9500
ROC AUC		0.9835	0.9817	0.9646	0.9896	0.8978	0.9957	0.9844
Balanced accuracy		0.9734	0.9856	0.9633	0.9720	0.8792	0.9720	0.9720
MCC		0.8750	0.9221	0.9341	0.9439	0.6964	0.9439	0.9463
Kappa		0.8711	0.9203	0.9341	0.9439	0.6928	0.9439	0.9457
Prevalence		0.0773	0.0773	0.0773	0.0773	0.0773	0.0773	0.0773
No information rate (NIR)		0.9227	0.9227	0.9227	0.9227	0.9227	0.9227	0.9227

The results presented in Table 6 demonstrate a consistent separation between models that encode nonlinear interactions and class boundaries (ensembles and artificial neural networks) and those that rely on a single global margin or local similarity (logistic regression, support vector machines and k-nearest neighbours). While the specificity levels are consistently high across the range of methods employed, reflecting the relative ease with which the majority class can be identified, the most significant disparities emerge in the retrieval of minority classes. The findings indicate that the ANN and the two ensemble methods (Random Forest, Gradient Boosting) achieve the optimal error trade-off by employing the F1($Y = 1$) and MCC metrics. This suggests that these methods not only effectively recover a substantial proportion of positives but also effectively mitigate false positives, thereby preserving decision utility in imbalanced settings.

A more thorough examination of the paired metrics illuminates the mechanisms underlying these gains. The Support Vector Machine (SVM) demonstrates the highest Balanced Accuracy, propelled by near-ceiling Recall ($Y = 1$) and robust Specificity. Nevertheless, its Precision ($Y = 1$) lags behind the ensembles and Artificial Neural Network (ANN), which consequently diminishes the F1 score. This pattern suggests that the SVM's linear margin though broad and sensitive admits more borderline positives, resulting in an inflation of recall at the expense of precision. In contrast, Random Forest and Gradient Boosting attain almost identical Recall($Y = 1$) with higher Precision($Y = 1$), implying superior class-boundary shaping and more selective positive predictions. The ANN combines high recall with competitive precision, producing the highest F1($Y = 1$) and MCC the latter of which summarises all four cells of the confusion matrix and thus provides a robust global check against spurious accuracy in imbalanced settings. Decision Tree is a close second, exhibiting marginally lower minority metrics but boasting a transparent structure and ease of explanation. Logistic regression provides a stable baseline with consistent but moderate F1($Y = 1$), while KNN underperforms in minority detection, consistent with its sensitivity to local density and class skew in tabular feature spaces.

Threshold-free discrimination (ROC–AUC) further corroborates these conclusions. The results of the study indicate that Gradient Boosting leads, followed by Random Forest and ANN, suggesting superior ranking quality across operating points. However, in practice, institutions act at a single operating threshold. Therefore, model selection should prioritise $F1(Y=1)$ (or cost-weighted utility) at clinically or economically relevant thresholds, rather than ROC–AUC alone. Two practical levers have been identified as having the potential to materially improve deployment outcomes: threshold optimization (e.g., selecting the point that maximises $F1(Y=1)$ or satisfies a mandated minimum Recall($Y=1$)) and probability calibration (Platt/Isotonic) to align scores with risk tolerances and downstream triage capacity. In view of the rarity of positives (Prevalence ≈ 0.0773), calibration is a means of avoiding overtriggering interventions while safeguarding sensitivity.

From a risk management perspective, the confusion matrices suggest the presence of distinct error profiles. Models with higher recall ($Y=1$) but somewhat lower precision (e.g., SVM) may be suitable for screening stages where the cost of a missed distressed firm is prohibitive, provided that downstream resources can absorb additional follow-up on false alarms. Conversely, the ensembles and ANNs have been shown to possess enhanced precision, rendering them particularly well-suited to prioritisation stages, where actionability and resource targeting are of the essence. The Decision Tree provides an interpretable alternative for auditability and policy communication.

Finally, the stability of these rankings under shifting class prevalence and macro conditions should be monitored. Periodic drift detection, recalibration, and retraining ideally with rolling windows or time aware validation can maintain reliability as financial cycles evolve. Incorporating exogenous signals (macro indicators, market volatility, sector stress, text based sentiment) and exploring cost-sensitive or recall constrained training objectives can further align model behavior with institutional risk appetite. In sum, the advanced metrics profile points to ANN and ensemble learners as the most effective for minority class detection without sacrificing overall correctness, while thresholding, calibration, and explainability close the final mile from technical performance to operational impact.

5 Discussion

The present study explores the potential of machine learning algorithms to predict financial crises in Indonesian businesses. This investigation utilises a substantial dataset encompassing financial and governance indicators, with the objective of developing a reliable predictive model. The findings indicate that sophisticated machine learning models, notably deep learning and ensemble methods such as gradient boosting, can enhance the precision of predictions concerning financial hardship. These models are especially useful for managing financial risk because they can find complex and non-linear patterns in financial data that other methods might miss. In the aftermath of the crisis, the practice of bankruptcy prediction has assumed increasing importance. This study introduces FS-Boosting, a novel, more accurate method for predicting corporate bankruptcy, the efficacy of which is proven through real data testing [51].

This study present to predict financial distress in Chinese construction companies listed on the stock exchange. To this end, the study compares the accuracy of machine learning models (CART, AdaBoost, and CUSBoost) with the traditional Z-Score model.

The findings indicate that machine learning algorithms, particularly CUSBoost, exhibit superior accuracy in predicting outcomes compared to the Z-Score method. This study is also the first to apply the CUSBoost algorithm in the context of financial distress prediction [52].

With the development of data mining, machine learning methods are increasingly used in financial prediction for stock selection [53]. That these models are consistently more accurate than statistical approaches [54, 55]. High feature correlation can destabilize classification algorithms. That tree models such as CART and random forest tend to ignore such correlation patterns [56, 57]. Financial problems in companies can often be seen in the way they manage assets and liquidity. Companies with good liquidity ratios such as cash, quick ratio, and current ratio, as well as strong asset coverage, tend to be more resilient to financial stress. While profitability is important to a company's success, this study shows that indicators such as gross profit and profit margin play little role in predicting financial distress. This means that even companies that look healthy in terms of profits can still run into trouble if their liquidity and asset management are not good.

Other factors such as board diversity and external conditions such as the corruption index also have an influence, although not dominant. Board diversity, for example, is considered to help risk management a little. However, since the effect is not large, further research is needed to understand the role of these factors more deeply. These non-financial matters are still important as a complement in looking at the overall financial stability of the company.

Research that supports the results of this study that has been conducted in Indonesia includes, Predicting financial distress is important in the midst of rapid business development in Indonesia. The Random Forest model with SMOTE technique proved to be the most accurate in detecting corporate financial risk. These results help choose the best method to prevent financial problems early on [58]. The COVID-19 pandemic has worsened the financial condition of many companies in Indonesia, making the prediction of financial distress very important. Compared to Naïve Bayes, the Random Forest model is proven to be more accurate in detecting potential bankruptcy in industrial companies. These results can be utilized as an early warning tool for management, investors, and regulators [59]. The pandemic reduced the financial performance of companies, especially in the telecommunications sector. Factors such as debt ratio, board of directors, and earnings management contribute to financial distress, while corporate strategy and firm size have no significant effect [60]. Neural networks, Random Forest, Deep Learning are used in research to minimize the level of prediction error [61].

In terms of prediction, machine learning models have proven to be superior to traditional methods. The use of deep learning models results in higher accuracy, while models such as random forest are still useful for providing a clearer understanding. For best results, a combination of both is recommended. Models also need to be updated regularly to stay relevant to changing market conditions. Going forward, combining financial data with unstructured data such as news or social media can improve prediction accuracy, while helping decision-makers understand risks more transparently through Explainable AI approaches.

6 Conclusion

The findings of this study demonstrate that contemporary machine-learning methodologies, notably ensemble models such as Gradient Boosting and Random Forest, in conjunction with a meticulously calibrated artificial neural network, are capable of attaining a commendable predictive capacity for the identification of corporate financial distress among Indonesian enterprises. Utilising a panel of 325 companies from 2013 to 2023, the models effectively capture nonlinear relationships in financial and governance indicators and achieve high Accuracy and F1 for the minority class under a leakage-safe evaluation protocol in which SMOTE is restricted to training folds and validation, test sets remain uncontaminated. The term ‘financial distress’ is defined as two consecutive years of negative equity, a transparent proxy for sustained solvency pressure and heightened reliance on debt. In practical terms, the findings suggest that firms aiming to reduce distress risk should prioritise liquidity management and asset coverage while exercising disciplined leverage.

However, it is important to note that several considerations must be taken into account when interpreting these results and formulating future research directions. The definition of the outcome, although clear and auditable, may not encompass all potential distress pathways, such as cash-flow shortfalls or covenant breaches. The sample, while sizable within the context, encompasses 325 firms over a period of eleven years and could be expanded to enhance external validity across sectors and market regimes. The feature set, with its emphasis on financial ratios, would benefit from augmentation with governance structures, regulatory shifts, sectoral exposures, event risk, and macroeconomic conditions, thereby enhancing its robustness. Addressing class imbalance with SMOTE has been shown to improve training, but creates a synthetic equilibrium that does not mirror real-world prevalence, underscoring the importance of evaluation on the untouched, imbalanced test distribution. In conclusion, despite the satisfactory performance of ensemble and deep models, they can be opaque to non-technical stakeholders. This emphasises the necessity for interpretability to be continued.

In order to enhance the decision-making capabilities of such systems in the future, there is a necessity to incorporate macro and sentiment signals, explore temporal and survival modelling that leverages sequence information, adopt cost-sensitive learning and calibrated decision thresholds aligned with institutional risk tolerances, and apply explainability methods such as permutation importance or SHAP to translate predictions into actionable drivers. It is imperative that routine model monitoring and recalibration be maintained in order to ensure reliability, particularly in the context of evolving market conditions. The evidence indicates that machine-learning approaches, when used in conjunction with rigorous evaluation and transparent explanation, can materially improve the early warning of corporate financial distress. This, in turn, supports risk management and policy efforts aimed at safeguarding employment and economic continuity.

Author contributions

F.T.K and D.F.S the fundamental body of the manuscript, conceived the instruments, and wrote the final paper. D.F.S analyzed and interpreted the data and was also a major contributor in writing the manuscript. A.F.I and D.F.S research design and the previous theoretical background. All authors read and approved the final manuscript.

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Data availability

The datasets generated by the survey research during and analyzed during the current study are available on the Indonesian Stock Exchange (IDX) <https://www.idx.co.id/en>, along with additional economic indicators provided by sources such as The data was extracted from the website <https://www.transparency.org/en/countries/indonesia>.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Anonymity and confidentiality

Not applicable.

Informed consent

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. El-Bannany M, Sreedharan M, M. A. A robust deep learning model for financial distress prediction. *Int J Adv Comput Sci Appl.* 2020;11 <https://doi.org/10.14569/IJACSA.2020.0110222>
2. Elhoseny M, Metawa N, Sztano G, El-hasnony IM. Deep learning-based model for financial distress prediction. *Ann Oper Res.* 2022. <https://doi.org/10.1007/s10479-022-04766-5>.
3. Li J, Wang C. A deep learning approach of financial distress recognition combining text. *Electron Res Arch.* 2023;31:4683–707. <https://doi.org/10.3934/era.2023240>.
4. Wu C, Jiang C, Wang Z, Ding Y. Predicting financial distress using current reports: A novel deep learning method based on user-response-guided attention. *Decis Support Syst.* 2024;179:114176. <https://doi.org/10.1016/j.dss.2024.114176>.
5. Irvan FM. Comparative analysis of machine learning and deep learning models integrated with Altman Z-Score for financial distress prediction in companies listed on the Indonesia Stock Exchange (IDX). *Ekombis Review Jurnal Ilmiah Ekonomi Dan Bisnis.* 2024. <https://doi.org/10.37676/ekombis.v12i2.5478>
6. Ramadani AW. Financial distress prediction: The role of financial ratio and firm size. *Jurnal Riset Akuntansi Kontemporer.* 2023. <https://doi.org/10.23969/jrak.v15i1.6428>
7. Rivanda AK, Afgani KF, Purbayati R, Marzuki MM. The effect of liquidity, leverage, operating capacity, profitability, and sales growth as predictors of financial distress: (Property, Real Estate, and Construction Services Companies Listed on the IDX). *J Integr Manag Stud.* 2023. <https://doi.org/10.58229/jims.v1i1.15>
8. Abdullah M, Mirosea N, Aswati WO, Santi N. Analysis of financial ratios to predict financial distress conditions of manufacturing companies listed on the Indonesian Stock Exchange. *Int J Prof Bus Rev.* 2023. <https://doi.org/10.26668/businessreview/2023.v8i7.3156>
9. Nurhayati N, Mufidah A, Kholidah AN. The determinants of financial distress of basic industry and chemical companies listed in Indonesia Stock Exchange. *Rev Manag Entrep.* 2018. <https://doi.org/10.37715/rme.v1i2.605>
10. Primawan A. Systematic literature review—determinants of corporate financial distress prediction. *Jurnal Akuntansi.* 2023. <https://doi.org/10.33369/jakuntansi.13.2.120-130>
11. Santoso N, Wibowo W. Financial distress prediction using linear discriminant analysis and support vector machine. *J Phys Conf Ser.* 2018. <https://doi.org/10.1088/1742-6596/979/1/012089>.
12. Lie D, Welly Y, Inrawan A, et al. Analysing the potential of bankruptcy using Altman Z-Score: A case study of PT. Garuda Indonesia. *Front Bus Econ.* 2022. <https://doi.org/10.56225/finbe.v1i1.80>
13. Sutra Tanjung PR, Anggraini D. Financial distress prediction of Lippo Group companies using Altman and Zmijewski models. *Saudi J Econ Financ.* 2020. <https://doi.org/10.36348/sjef.2020.v04i01.001>
14. Paranita ES, Andri. Financial distress during the COVID-19 pandemic: Altman and Springate model prediction. *SFR.* 2024. <https://doi.org/10.59762/sfr794324261120240118154533>
15. Handayati P, Izzalqurny TR, Fauzan S, Shobah N. The phenomenon of financial distress of manufacturing companies in Indonesia during the Covid-19 pandemic. *Int J Res Bus Soc Sci* (2147–4478). 2022. <https://doi.org/10.20525/ijrbs.v11i9.2205>
16. Altman EI. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *J Finance.* 1974;29(1):3–12. <https://doi.org/10.1111/j.1540-6261.1974.tb00057.x>.
17. Samitas A, Kampouris E, Kenourgios D. Machine learning as an early warning system to predict financial crisis. *Int Rev Financ Anal.* 2020;71:101507.
18. Kim S, Ku S, Chang W, Song JW. Predicting the direction of US stock prices using effective transfer entropy and machine learning techniques. *IEEE Access.* 2020;8:111660–82.
19. Zeng S, Li Y, Yang W, Li Y. A financial distress prediction model based on sparse algorithm and support vector machine. *Math Prob Eng.* 2020;2020:1–11.
20. Elhoseny M, Metawa N, Sztano G, et al. Deep learning-based model for financial distress prediction. *Ann Oper Res.* 2025;345:885–907. <https://doi.org/10.1007/s10479-022-04766-5>.

21. Ben Jabeur S, Serret V. Bankruptcy prediction using fuzzy convolutional neural networks. *Res Int Bus Finance*. 2023;64. <https://doi.org/10.1016/j.ribaf.2022.101844>.
22. Chaloumis C, Eriotis N. A historical analysis of the banking system and its impact on Greek economy. *Edelweiss Appl Sci Technol*. 2024;8(6):1598–617. <https://doi.org/10.55214/25768484.v8i6.2282>.
23. Halim Z, Shuhidan SM, Sanusi ZM. Corporation financial distress prediction with deep learning: analysis of public listed companies in Malaysia. *Bus Process Manag J*. 2021;27(4):1163–78. <https://doi.org/10.1108/BPMJ-06-2020-0273>.
24. Oribel T, Hanggraeni D. An application of machine learning in financial distress prediction cases in Indonesia. *Int J Bus Technol Manag*. 2021;3(2):98–110.
25. Metwally ABM, Yasser MM, Ahmed EA, Ali MAS. Financial and economic determinants of banks financial distress in MENA region. *Economies*. 2025;13(2):56. <https://doi.org/10.3390/economies13020056>.
26. Ashraf S, Félix EG, Serrasqueiro Z. Do traditional financial distress prediction models predict the early warning signs of financial distress? *J Risk Financ Manag*. 2019;12(2):55. <https://doi.org/10.3390/jrfm12020055>.
27. Pham Vo Ninh B, Do Thanh T, Vo Hong D. Financial distress and bankruptcy prediction: an appropriate model for listed firms in Vietnam. *Econ Syst*. 2018;42(4):616–624. <https://doi.org/10.1016/j.ecosys.2018.05.002>.
28. Halteh K, Alkhoury R, Ziadat SA, Gepp A, Kumar K. Using machine learning techniques to assess the financial impact of the COVID-19 pandemic on the global aviation industry. *Transp Res Interdiscip Perspect*. 2024;24:101043. <https://doi.org/10.1016/j.trip.2024.101043>.
29. Jin B, Xu X. Carbon emission allowance price forecasting for China Guangdong carbon emission exchange via the neural network. *Global Finance Rev*. 2024;6(1):3491. <https://doi.org/10.18282/gfr.v6i1.3491>.
30. Khemakhem S, Ben Said F, Boujelbene Y. Credit risk assessment for unbalanced datasets based on data mining, artificial neural network and support vector machines. *J Model Manag*. 2018;13(4):932–51. <https://doi.org/10.1108/JM2-01-2017-0002>.
31. Liu S, Wang X, Li C, Zhang Y. Financial modeling of common institutional ownership's governance effect on corporate leverage manipulation. *Mathematics*. 2024;13(1):1–25. <https://doi.org/10.3390/math13010093>.
32. Jin B, Xu X. Pre-owned housing price index forecasts using Gaussian process regressions. *J Modelling Manag*. 2024;19(6):1927–58. <https://doi.org/10.1108/JM2-12-2023-0315>.
33. Kuiziniene D, Krilavičius T, Damaševičius R, Maskeliūnas R. Systematic review of financial distress identification using artificial intelligence methods. *Appl Artif Intell*. 2022;36(1):2138124. <https://doi.org/10.1080/08839514.2022.2138124>.
34. Ntoun L, Santos de Oliveira HM, Sousa BMF, Pimentel LM, Bastos SA. Are family firms financially healthier than non-family firms? *J Risk Financial Manag*. 2019;13(1):5. <https://doi.org/10.3390/jrfm13010005>.
35. Pech M, Prazakova J, Pechova L. The evaluation of the success rate of corporate failure prediction in a five-year period. *J Competitiveness*. 2020;12(1):108–24. <https://doi.org/10.7441/joc.2020.01.07>.
36. Vukčević M, Lakičević M, Melović B, Backović T, Dudić B. Modern models for predicting bankruptcy to detect early signals of business failure: evidence from Montenegro. *PLoS ONE*. 2024;19(5):1–19. <https://doi.org/10.1371/journal.pone.0303793>.
37. Gaur V, Kesavan S. The effects of firm size and sales growth rate on inventory turnover performance in the U.S. retail sector. *Retail Supply Chain Manag*. 2015;233:25–52. https://doi.org/10.1007/978-1-4899-7562-1_3.
38. Onchangwa GA, Memba FS, Nasieku TM. The relationship between receivables management and financial distress of non-financial firms listed at Nairobi Securities Exchange in Kenya. *J Multidiscip Eng Sci Technol (JMEST)*. 2018;5(10):2458–9403.
39. Sun J, Shang Z, Li H. Imbalance-oriented SVM methods for financial distress prediction: a comparative study among the new SB-SVM-ensemble method and traditional methods. *J Oper Res Soc*. 2014;65(12):1905–19. <https://doi.org/10.1057/jors.2013.117>.
40. Dirman A. Financial distress: the impacts of profitability, liquidity, leverage, firm size, and free cash flow. *Int J Bus Econ Law*. 2020;22(1):17–25.
41. Vo DH, Pham BN, Ho CM, McAleer M. Corporate financial distress of industry level listings in Vietnam. *J Risk Financial Manag*. 2019;12(4):155. <https://doi.org/10.3390/jrfm12040155>.
42. Rao K, Tilt C. Board composition and corporate social responsibility: the role of diversity, gender, strategy and decision making. *J Bus Ethics*. 2016;138(2):327–47. <https://doi.org/10.1007/s10551-015-2613-5>.
43. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: Synthetic minority over-sampling technique. *J Artif Intell Res*. 2002;16:321–57. <https://doi.org/10.1613/jair.953>.
44. Quinlan JR. Induction of decision trees. *Mach Learn*. 1986;1:81–106.
45. Breiman L. Random forests. *Mach Learn*. 2001;45:5–32.
46. Friedman JH. Greedy function approximation: a gradient boosting machine. *Ann Stat*. 2001;29:1189–232.
47. Cover T, Hart P. Nearest neighbor pattern classification. *IEEE Trans Inf Theory*. 1967;13:21–7.
48. Breiman L. Random Forests. *Mach Learn*. 2001;45:5–32. <https://doi.org/10.1023/A:1010933404324>.
49. Goodfellow I, Bengio Y, Courville A. Deep learning. MIT Press. 2016.
50. Beaver WH. Financial ratios as predictors of failure. *J Account Res*. 1966;4(1):71–111.
51. Rahman MJ, Zhu H. Predicting financial distress using machine learning approaches: evidence from China. *J Contemp Account Econ*. 2024;20(1):100403. <https://doi.org/10.1016/j.jcae.2024.100403>.
52. Albanis G, Batchelor R. Combining heterogeneous classifiers for stock selection. *Intell Syst Account Financ Manag*. 2007;15(1–2):1–21.
53. De Andrés J, Lorca P, de Cos Juez FJ, Sánchez-Lasheras F. Bankruptcy forecasting: A hybrid approach using Fuzzy c-means clustering and Multivariate Adaptive Regression Splines (MARS). *Expert Syst Appl*. 2011;38(3):1866–75.
54. Karas M, Reznakova M. Predicting the bankruptcy of construction companies: a CART-based model. *Eng Econ*. 2017;28(2):145–54.
55. Toloşi L, Lengauer T. Classification with correlated features: unreliability of feature ranking and solutions. *Bioinformatics*. 2011;27(14):1986–94.
56. Rabinowicz A, Rosset S. Tree-based models for correlated data. *J Mach Learn Res*. 2022;23(1):11802–32.
57. Kristanti FT, Febrianti MY, Salim DF, Riyadh HA, Beshr BAH. Predicting Financial Distress in Indonesian Companies using Machine Learning. *Eng Technol Appl Sci Res*. [Internet]. 2024 Dec. 2 [cited 2025 Jul. 17];14(6):17644–9. Available from: <http://etasr.com/index.php/ETASR/article/view/8520>

58. Elsamayori D, Kristanti FT. Unveiling Financial Distress: A Comparative Analysis of Random Forest and Naïve Bayes Models. In: 2025 IEEE international conference on interdisciplinary approaches in technology and management for social innovation (IATMSI). Vol. 3, IEEE; 2025. pp 1–6.
59. Kristanti FT, Pancawitri S. Some factors affecting financial distress in telecommunication companies in Southeast Asia. BTP [Internet]. 2024 Mar. 27 [cited 2025 Jul. 17];25(1):190–199. Available from: <https://journals.vilniustech.lt/index.php/BTP/article/view/20018>
60. Inam SA, Khan AA, Ahmed N, et al. A novel deep learning approach for investigating liquid fuel injection in combustion system. *Discov Artif Intell*. 2025;5:32. <https://doi.org/10.1007/s44163-025-00248-2>.
61. Inam SA, Khan AA, Mazhar T, et al. PR-FCNN: a data-driven hybrid approach for predicting PM2.5 concentration. *Discov Artif Intell*. 2024;4:75. <https://doi.org/10.1007/s44163-024-00184-7>.

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